Conservative-First Release under Finite Verifiers in Generated Agent Workflows

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Abstract

When a finite test set—a verifier—decides which generated workflow changes an enterprise agent may release, optimizing past acceptance can overfit the verifier: in our discovery phase, an exact subset search that broke test-pass ties toward fewer rules deleted rules whose value shows only on scenario-disjoint hidden cases. We study *Bundle-then-Subset* (BTS), a conservative-first release rule: release a verifier-accepted bundle unchanged, search subsets only after rejection, and otherwise abstain. In a confirmation preregistered against Bundle and Direct (eight synthetic organizations, two models), BTS ties exhaustive subset search, a comparison not preregistered (DeepSeek: 0.704 SC-AUC, identical SC-VAC in every unit, 88.6% fewer verifier evaluations than plain enumeration). Two stress arms with designed verifier blind spots show the failure: a fewer-rules tie-break deletes every verifier-invisible rule from every accepted bundle, costing hidden coverage or, where surviving rules relied on rule order, causing policy violations. A post-hoc control that breaks ties toward more rules avoids this as well as BTS or better under the full verifier, where BTS only saves verifier evaluations. Neither protects against verifier false acceptances, which BTS inherits.

1 Introduction

An enterprise agent must learn more than generic tool use: which source is authoritative, which policy version is current, how local tools compose, and when a human must approve an action. We study whether this organization layer can emerge in a minimal harness while the base model remains frozen. Improvement edits external workflows, not weights; its input is chronological organizational evidence rather than gradients or scalar reward.

The release step is where this study lives. A generated workflow cannot be trusted on sight, so before release it must pass a *verifier*: a finite panel of test cases—here, eight per organization—that executes the workflow and checks decisions, process obligations, and policy edges. The panel is a sample of the behavior the organization will actually see, so a workflow's verifier score measures only the behavior the panel happens to exercise, and any optimization that pushes past "the tests pass" optimizes an incomplete proxy. The sharpest instance is subset minimization: if a rule—an approval requirement on high-value requests—has no exercising case in the panel, deleting it cannot change the score, and a fewer-rules preference, standard in test-based minimization [14, 23], will then delete it every time. The removed obligation stays invisible until deployment traffic exercises it. This is the release-time analogue of test-suite overfitting in program repair [17]: the verifier is overfit as a model overfits a validation set, except that what is overfit is the released artifact. Minimizing an accepted artifact is not contrived: wherever an optimizer prices artifact size or inference cost, as in cost-aware workflow search, cloud-edge offloading [9], and prompt or context compression, ties

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at equal score go to the smaller artifact, and context optimization for agents shows an analogous brevity bias [25]. Breaking ties toward more rules is an equally valid fix, and we test it (Section 4).

Harness Continual Learning (HCL) formalizes a persistent, jointly versioned task interface, memory, capability map, and router around a frozen foundation model, with a retention-aware evaluator mediating proposal-to-commit transitions [11]. We isolate a narrower release-granularity question: when one proposed workflow is rule-decomposable and judged by a finite verifier, should a controller preserve an accepted bundle or continue searching its subsets? We hold candidate generation, representation, evidence, model, and verifier fixed and vary only commitment within that candidate. Because checkpoints independently regenerate workflows, we study evidence-conditioned harness acquisition and release rather than retention across committed versions. BTS is therefore a release component, not a complete continual-learning system: a stateful pipeline in the HCL sense still needs a commitment rule at every proposal-to-commit transition, which BTS could fill; evaluating it with a stateful updater is future work (Section 6).

Generated improvements require a release controller that the improving component cannot rewrite. This study follows a discovery–confirmation sequence. A post-hoc diagnosis of the preregistered v0.2 comparison found that exact subset selection both rescued rejected bundles and, for some verifier-accepted bundles, used a fewer-rules tie-break to select verifier-equivalent subsets that removed rules useful on scenario-disjoint hidden cases. This failure suggests a narrow principle: against a fixed finite verifier, further optimization of an accepted artifact must not trade away rules the verifier cannot see, and terminating optimization at acceptance is the simplest way to ensure it. It does not imply that an accepted artifact is truly safe. A verifier-defined selection objective, including its parsimony tie-break, determines which behavior is preserved.

We make three contributions. (1) We trace verifier-induced over-pruning of accepted generated workflows to the fewer-rules tie-break and propose BTS, a failure-driven operator that restricts fine-grained search to recovery after rejection; a post-hoc control shows that breaking ties toward more rules also avoids the over-pruning, equally well or better under the full verifier, so BTS's remaining edge there is fewer verifier evaluations. (2) In the fresh-domain confirmation, BTS ties exhaustive Safe-Subset (0.704 SC-AUC, identical SC-VAC in every unit), so v0.3 shows no accuracy or safety advantage over exact search; what BTS changes is cost and what it releases: 88.6% fewer verifier artifact evaluations than plain enumeration and no accepted-bundle pruning. Its preregistered H1 gain over Bundle (+0.076) is coverage recovered from rejected bundles, since BTS cannot do worse on SC-VAC than an abstaining Bundle. (3) Two prospective stress arms with different designed verifier blind spots, each run on both models, show both sides of the trade on largely shared candidates. When ties favor fewer rules, subset search deletes every rule that no verifier case matches from every accepted bundle; the resulting exposure is benign abstention or rule-order policy violation depending on whether the surviving rules guard the blind behavior, and the frozen benign-shape point prediction failed in three of the four arms (Section 4). Conservative-first release preserves the accepted bundle in every unit, and what it inherits instead are false acceptances wherever the verifier is too weak to reject.

2 Harness and conservative-first release

At checkpoint b, a fixed model receives evidence prefix $E_{\leq b}$, declared tool names, and a task-family input schema. It emits one provenance-backed workflow in a restricted DSL of at most eight conditional rules. Rules may call declared tools, request approval, escalate, respond, or abstain. Arbitrary code, new permissions, editable graders, and weight updates are excluded. A second call may only repair a statically invalid artifact; behavioral feedback never changes the shared candidate.

Each checkpoint is an independent snapshot: W_b is regenerated from $E_{\leq b}$ and does not receive the previously released artifact B_{b-1}. The acquisition curve therefore does not represent one stateful $H_b \to H_{b+1}$ trajectory. BTS preserves rules within the same candidate, not behavior across sequentially committed harness versions.

Direct activates every statically valid candidate. *Bundle* activates the complete candidate only if an eight-case verifier observes zero policy/grounding failures and at least one automated pass. *Safe-Subset* enumerates all nonempty rule subsets and maximizes verifier automated passes under the same safety constraints, breaking ties toward fewer rules and then toward the first subset in enumeration order. All gates receive the same candidate.

For candidate W_b, let $V(W_b)$ mean that the complete candidate passes the Bundle contract, and let S_b^* be Safe-Subset's best nonempty subset. Bundle-then-Subset (BTS) selects

$$B_b = \begin{cases} W_b, & V(W_b), \\ S_b^*, & \neg V(W_b) \ \wedge \ S_b^* \text{ exists}, \\ \emptyset, & \text{otherwise.} \end{cases} \tag{1}$$

The over-pruning failure this prevents is structural, not incidental:

Proposition 1 (blind-stratum pruning) *Let V be a finite verifier on which abstention earns no automated pass, and let W be a V-accepted candidate containing a nonempty rule set R_s none of whose rules matches any V-exercised input (no verifier case satisfies their conditions; the stratum-s obligation they encode has no exercising case in the panel). Because a non-matching rule never fires under first-match execution, deleting the R_s rules from any subset $T \subseteq W$ of the accepted W leaves every V-exercised outcome unchanged; whenever $T \setminus R_s \neq \emptyset$, that smaller subset ties T on the verifier objective with strictly fewer rules, and a subset containing only R_s rules produces no automated verifier pass, which the gate requires. Hence Safe-Subset's selected artifact, when one exists, contains no rule of R_s. BTS releases W unchanged if and only if $V(W)$, so on the accepted path it preserves R_s.*

The proof is contained in the statement: the objective reads only V-exercised outcomes, non-matching rules cannot alter them in any subset of an accepted W, fewer rules wins the tie-break (the enumeration-order rule only chooses among equal-size subsets), and the at-least-one-pass requirement excludes all-R_s subsets. The effect is not new: reduced-error pruning removes every subtree that receives no pruning examples because ties go to the smaller tree [5], and minimizing an artifact to 1-minimality against a test predicate likewise discards whatever the tests never exercise [6, 14, 23]. Proposition 1 is an instance of it for release gates over generated rule bundles, where the deleted behavior can be a policy obligation, and it is deliberately narrow: it covers only rules that match no verifier input and only selectors whose ties favor fewer rules (ties toward more rules keep every blind rule; Section 4), and says nothing about rules that fire on verifier cases without adding passes, which can be pruned too. Nor does it say that the accepted W is safe (the false-acceptance boundary), when parsimony *helps* (the thin-evidence stratum, where pruning discards falsely accepted behavior that Bundle and BTS retain), or what happens when a blind stratum is partially exercised rather than absent.

This guarantee is path-specific: BTS prevents accepted-bundle pruning; it does not certify acceptance. If V falsely accepts a hidden-unsafe bundle, BTS preserves that error by construction, whereas Safe-Subset could incidentally remove the offending rule. BTS is therefore not theoretically weakly dominant. Promotion is deterministic and external; the model cannot see expected actions, gate outcomes, or hidden cases.

3 Benchmark, metrics, and freeze

The preregistered v0.2 comparison is our discovery phase. Exact subset release failed to beat Bundle (-0.005 SC-AUC). A post-hoc decomposition identified 46 rescues offset by the 94 of 219 accepted-bundle pruning events that lowered hidden coverage, all in the 4 procurement organizations, and the 4 support organizations' 71 prunings cost nothing. BTS was then reconstructed post-hoc from frozen aggregates (+0.028 [+0.018, +0.037] over Bundle) and preregistered before v0.3 hidden results were unlocked, with contrasts against Bundle (H1) and Direct (H2); the BTS-versus-Safe-Subset contrast that motivated it was not preregistered, and it is null in v0.3.

The failure in one unit. One discovery-phase unit makes the mechanism concrete. At checkpoint 16 of one v0.2 procurement organization, the updater emitted a four-rule workflow: escalate unapproved vendors, execute low-value approved purchases, and route mid- and high-value purchases to approval. The complete bundle passed all eight verifier cases and covered $12/12$ hidden tasks. Exact subset search evaluated all 15 nonempty subsets and found a three-rule subset that also passed all eight cases, so the fewer-rules tie-break dropped the high-value approval-routing rule; hidden coverage fell to $10/12$ because high-value purchases now matched no rule. Independently regenerated candidates in this organization lose the same behavior across seeds and later checkpoints: the high-value routing rule, under varying rule names, with the identical $12/12$ to $10/12$ drop each time.

Table 1: v0.3 frozen hidden results; final VAC 1.000 is omitted. Verifier artifact evaluations count each complete candidate or rule subset tested by a gate.

Gate	SC AUC ↑	Raw AUC ↑	Viol. AUC ↓	Verifier artifact evals ↓
Direct	0.628	**0.726**	0.046	—
Bundle	0.628	0.628	**0.000**	**381**
Safe-Subset	**0.704**	0.704	**0.000**	9,315
Bundle-then-Subset	**0.704**	0.704	**0.000**	1,063

Nothing unsafe occurs: the pruned workflow simply stops automating behavior its test set never probes. This is verifier overfitting through a parsimony preference; BTS removes it by construction, as does a tie-break toward more rules.

Before unlocking v0.3, we froze eight organizations, four from each of two task-family templates (expense reimbursement, access provisioning) that vary names, thresholds, and tools. Each has 32 items (6 documents, 18 messages, 4 tool schemas, 4 corrections), an eight-case verifier (3 decision, 2 workflow, 3 policy-edge), and 12–13 scenario-disjoint hidden cases. The final evidence multiset is fixed while a current action threshold and its approval rule are separated by 0–3 eight-item batches. Checkpoints are at 0, 8, 16, 24, and 32 items.

Verified automation coverage (VAC) requires simultaneous functional, policy, and grounding passes. An artifact is deployable only if every hidden trace at a checkpoint passes policy and grounding:

$$\text{SC-VAC}(b) = \text{VAC}(b)\,\mathrm{I}[\forall t,\ P_t \wedge G_t]. \tag{2}$$

The primary metric is normalized trapezoidal SC-VAC AUC. Raw VAC AUC, final VAC, violation-exposure AUC, gate route, and a frozen 13-class outcome taxonomy are mandatory. The taxonomy distinguishes useful action and human routing from abstention, policy failure, functional failure, and grounding failure; classes are mutually exclusive and aggregated before persistence. All gates reach final VAC 1.000 in the DeepSeek arms (0.990 under Gemini, reflecting its one rejected generation), so the comparison concerns acquisition speed rather than terminal capability.

The preregistered v0.3 matrix has 8 organizations $\times$ 4 gaps $\times$ 5 checkpoints $\times$ 3 seeds = 480 units (one generation per seed, none at checkpoint 0; identical prompts share cached candidates) and 24,000 gate-task evaluations. DeepSeek V4 Pro runs in JSON mode at temperature zero with reasoning and provider retries disabled. The organization, not the workflow unit, is the cluster ($n = 8$; as parameterized instances, they measure sensitivity to surface parameters, not between-organization variation); percentile 95% intervals use 10,000 organization-cluster bootstrap resamples. All units, manifests, artifacts, frozen hashes, and outcome counts passed audit; no task-level hidden traces were retained. Provider cost was $0.5082.

4 BTS matches exact search; the tie-break causes over-pruning

BTS ties exact subset search with far less verifier work. Safe-Subset and BTS both reach 0.704 SC-AUC in v0.3, and their SC-VAC is identical in all 480 units, so v0.3 establishes no accuracy or safety advantage over exact subset search. BTS instead uses 1,063 versus 9,315 verifier artifact evaluations (88.6% fewer than plain enumeration; Appendix E) and makes accepted-bundle pruning impossible, whereas Safe-Subset prunes 112 accepted bundles, removals that happen not to alter hidden coverage here. With at most eight rules, this is evidence about search structure, not production-scale runtime.

The gain over Bundle is recovered coverage. The preregistered H1 contrast, BTS minus Bundle, is +0.076 [+0.040, +0.111]. It is not a better decision on shared ground: both gates release an accepted bundle unchanged and Bundle abstains on a rejected one, so BTS is non-worse by construction; its SC-VAC equals Bundle's in all 311 accepted units, exceeds it in all 70 subset-recovery units, and is never lower. BTS preserves the full bundle in 81.0% of nonzero units, recovers a subset in 18.2%, and abstains in 0.8%. Relative to Direct, BTS's preregistered H2 SC-AUC gain equals H1's, because Direct and Bundle tie on SC-AUC here; as preregistered companions, it changes raw AUC by -0.022 [-0.035, -0.010] while reducing violation-exposure AUC by 0.046 [0.025, 0.068].

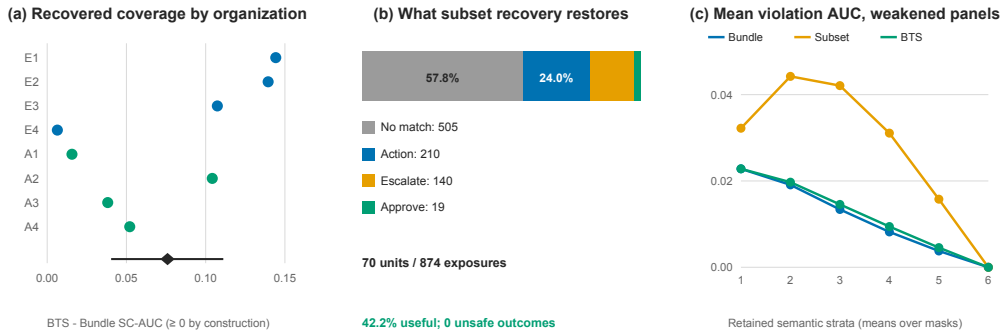


Figure 1: Post-hoc diagnostics of the frozen v0.3 run. (a) BTS-minus-Bundle organization effects (E: expense; A: access), non-negative by construction, so they measure recovered coverage; the diamond is the cluster mean and bootstrap interval. (b) Aggregate outcomes in 70 subset-recovery units. (c) Mean violation AUC over exhaustive masks at each number of retained semantic verifier strata; per mask, BTS is lower/equal/higher in 24/17/22 of these 63 masks (Appendix D).

A second model replicates the ordering. The v0.3 freeze prespecified a conditional cross-model arm (Gemini 3.7 Flash). All 480 units ran with benchmark, matrix, seeds, and analysis byte-identical to v0.3 (successor protocols: Appendix B); $383/384$ eligible outputs were valid. The DeepSeek gate ordering replicates: per-gate SC-AUC is 0.627 (Direct), 0.627 (Bundle), 0.705 (Safe-Subset), 0.705 (BTS), against DeepSeek's $0.628/0.628/0.704/0.704$, and H1 is $+0.078$ $[0.067,\ 0.092]$ against DeepSeek's +0.076; the mandatory model-stratified pooled H1 over all 960 units is $+0.077$ $[0.058,\ 0.096]$. The two models make the same bundle accept/reject decision in 348 of the 380 matched units where both generated a candidate (post-hoc, zero model calls); the per-gate means agree to within 0.002 partly through offsetting per-unit differences, not because every unit agrees. Thinking level and serving stack also differ, so this is a cross-configuration replication, not a pure model swap.

Recovery anatomy. Across 70 subset-recovery units (874 hidden-task exposures), BTS produces 210 successful actions, 140 correct escalations, 19 correct approvals, and 505 no-rule-match outcomes, so 42.2% regain correct action or human routing, with no incorrect or unsafe outcome. All but two rescues occur at the first nonzero checkpoint: the gain is early acquisition, not terminal capability.

The verifier sets the safety ceiling. The full eight-case verifier exactly separates all 381 observed nonempty bundles: 311 accepts are hidden-deployable and 70 rejects are not, with no false acceptances or rejections; the Gemini arm's 383 bundles separate just as exactly (300 and 83). This is a synthetic-benchmark diagnostic, not evidence that eight tests suffice: verifier and hidden scenarios share a typed organization specification, and both labels are coarse binaries. We therefore exhaustively replay all 255 nonempty verifier subsets, post-hoc, with stored candidates and zero model calls. Per mask, BTS's mean SC-AUC is never below Safe-Subset's (0/21/234 masks lower/equal/higher, although 140 of 24,480 individual seed series are lower), but its violation exposure is lower in 112 masks, equal in 67, and higher in 76, nearly even over the 63 semantic-strata masks of Figure 1(c) (24/17/22), and the split turns on whether the panel keeps the escalation case D8 and the special-approval case D3 (Appendix D). The coverage edge comes from the tie-break: Max-Subset (defined below) beats/ties/trails BTS's mean SC-AUC in 168/66/21 masks (trailing by at most 0.0009), with lower/equal/higher violation exposure in 98/117/40. Omitting D3 alone creates 10 Bundle false acceptances (14.3% of hidden-nondeployable bundles); BTS inherits them and reaches 0.0022 violation AUC, versus Safe-Subset's 0.00087. Omitting the escalation case alone leaves full-bundle separation intact but makes subset recovery unsafe (BTS 0.0047; Safe-Subset 0.0733). BTS is not verifier-independent.

A designed blind spot isolates the mechanism. Because v0.3's verifier proved representative enough that its 112 accepted-bundle prunings left hidden coverage unchanged, a prospective stress arm (v0.5) designs the blind spot: six fresh organizations whose eight-case development panel contains no approval-outcome case, while their hidden panels are approval-rich; DSL, prompts, gates,

metrics, and the v0.3 model configuration are unchanged (6 organizations $\times$ 5 checkpoints $\times$ 3 seeds $= 90$ units; predictions and protocol frozen before the hidden matrix, after a two-organization development check; Appendix A). **P1** and **P2** are design-validation checks — given the proposition, they are arithmetic consequences once candidates encode the blind obligation — and both held: given the premise that every generated candidate encodes the approval obligation ($72/72$), exact subset search pruned the approval rules from every accepted bundle ($66/66$; P1), and Safe-Subset's hidden coverage fell below Bundle's and BTS's (P2). The empirical content lies in the premises holding for generated candidates at all, and in what follows. At the *evidence-complete* checkpoints (16–32, where all $54/54$ accepted bundles pass the full development panel; a post-hoc stratum for v0.5, first named in the v0.5b protocol), the per-unit coverage loss equals the blind approval-task share *exactly* in $54/54$ units, a six-organization mean of 0.48 that is a design constant (0.46 expense, 0.50 access), so we report no interval. BTS equalled Bundle in every accepted unit and additionally recovered rejected units (overall SC-AUC 0.65 versus 0.64; Safe-Subset 0.36; Figure 2). **P3** (the pruning harm is lost coverage, not unsafe action) held too: approval-outcome tasks fall to no-rule-match abstention, and no violation is attributable to pruning.

The thin-evidence checkpoint adds an *observed, unpredicted* boundary. From eight evidence items the blind panel accepted twelve weak bundles, ten of them falsely; those ten commit twenty hidden policy violations that Bundle and BTS inherit while Safe-Subset incidentally avoids them by pruning the offending rules, exactly the asymmetry anticipated in Section 2.

A second blind spot flips the failure shape. To test whether the mechanism is specific to the approval stratum, a second frozen arm (v0.5b) blinds the development panel to a different behavior: it contains both approval cases but no ineligible case, so the eligibility/escalation rule is the invisible one, and hidden panels carry every ineligible case (three per organization). Its organizations, evidence, prompts, and model configuration are byte-identical to v0.5, so the response cache gives both arms the *same* candidates in 68 of 72 cases (one seed series was regenerated), a paired comparison (Appendix A). The pruning half repeats exactly: every candidate encoded the eligibility obligation ($72/72$), subset search pruned the escalation rule from every accepted bundle that carried it ($56/56$), and the per-unit coverage loss again equals the blind share exactly ($54/54$; mean 0.24). The exposure shape, however, flips, and our frozen point prediction fails: P3' predicted the benign shape at evidence-complete checkpoints, while committing to report whichever branch occurred. Twenty-six of the 54 evidence-complete accepted units relied on escalate-first rule order rather than an explicit eligibility guard in their tier rules; with the escalation rule pruned, those artifacts auto-execute ineligible requests, and Safe-Subset becomes the violating gate (52 hidden policy violations at the evidence-complete checkpoints) while Bundle and BTS commit none. The same exact coverage loss splits into benign abstention (28 units) and unsafe action (26 units) depending on a rule-order dependence that no coverage metric distinguishes; the safety-constrained metric does. Table 2 aligns the two arms.

Both designs were repeated with Gemini 3.7 Flash under jointly frozen successor protocols (Appendix B). The mechanism transfers exactly: every candidate encodes the blind obligation, subset search prunes it from every accepted bundle, and coverage loss equals the blind share in $54/54$ evidence-complete units in both arms (Table 2). The violation channel is candidate-dependent: Gemini relied on rule order in both arms ($16/54$ escalation-blind units, 32 violations; $5/54$ approval-blind units, 30 violations), so the benign-shape point prediction fails in both (for the escalation-blind arm the jointly frozen protocol text states only the disjunctive P3', which held, while the arm's freeze record also lists the benign point prediction; Appendix B). What holds across models is the rule: pruning is benign exactly where the surviving rules guard the blind behavior, at evidence-complete checkpoints violations appear under Safe-Subset only, never under Bundle or BTS, and BTS preserved every accepted bundle in all four arms.

The failure is the tie-break. A post-hoc control isolates the damaging ingredient. *Max-Subset* runs the identical exact search—same safety constraints, same pass maximization—except that ties break toward *more* rules. Replaying every stored candidate through the frozen gates (zero model calls; both models, the confirmatory arms, and all four stress arms), Max-Subset retains the complete bundle in $856/856$ accepted-bundle units and adds no violation of its own (its only violations are the inherited false acceptances below); its SC-AUC equals Safe-Subset's where nothing was ever lost (0.704 and 0.705 in the two confirmatory arms) and exceeds BTS's on the escalation-blind rejected path, where its maximal rescued subsets carry more coverage than BTS's minimal ones

Table 2: **In all four arms, Safe-Subset prunes the blind rule from every accepted bundle; BTS never prunes any.** Two designed verifier blind spots, each over two models' candidates (six organizations; first two rows over all nonzero checkpoints, the rest evidence-complete unless noted; a model's two arms share 68/72 cached candidates under DeepSeek, 72/72 under Gemini). Coverage cost is the organization mean of per-unit accepted-bundle minus Safe-Subset hidden coverage; it agrees across models because the loss equals the designed blind share in every unit, a design constant reported without an interval, while violations depend on each model's rule order.

	DeepSeek V4 Pro		Gemini 3.7 Flash	
	Approval-blind	Escalation-blind	Approval-blind	Escalation-blind
Candidates encoding blind obligation	72/72	72/72	72/72	72/72
Accepted bundles, blind rule pruned	66/66	56/56	66/66	57/57
Coverage loss = blind share	54/54	54/54	54/54	54/54
Coverage cost (= blind share, mean)	0.48	0.24	0.48	0.24
Safe-Subset hidden violations	0	52 (26 units)	30 (5 units)	32 (16 units)
Bundle/BTS hidden violations	20 (thin-ev.)	0	18 (thin-ev.)	0
BTS equals accepted Bundle	every unit	every unit	every unit	every unit

(0.732 versus 0.688 under DeepSeek; 0.707 versus 0.675 under Gemini); under approval blindness it ties or nearly ties (0.657 versus 0.653; 0.647 versus 0.647). The over-pruning failure is therefore attributable to the parsimony tie-break, not to subset search: deleting verifier-invisible rules only ever ties, so the tie-break direction alone decides their fate. What conservative-first release adds over reversing the tie-break is not accuracy but less verifier work: BTS releases the same artifact on every accepted bundle without searching subsets, with 88.6% (DeepSeek) and 81.5% (Gemini) fewer verifier artifact evaluations than plain enumeration, but at most 50.8% and 18.7% fewer than a Max-Subset search that stops once the complete bundle passes every case (Appendix E). Its own recovery path still uses the fewer-rules tie-break; post-hoc variants that recover by descending from the bundle match or beat it under the full verifier with fewer evaluations (Appendix E), so frozen BTS is not the best point in this design space. Like Bundle and BTS, Max-Subset inherits the thin-evidence false acceptances (20 and 18 violations); reversing the tie-break is no protection against a verifier too weak to reject.

A designed blind spot estimates no prevalence. The pair shows that when release tests cannot see a behavior, a fewer-rules search deletes it from every accepted candidate, and *which* operator then looks safe depends on where the verifier is blind: in one arm conservative-first release inherits false acceptances that pruning would have removed; in the other, pruning manufactures the violations.

Retaining a bundle has a cost. On the 311 v0.3 accepted units, Bundle and BTS release 1,453 rules versus Safe-Subset's 1,326 (+9.6%) and make 2.612 versus 2.536 rule-condition evaluations per hidden task (+3.0%). This v0.3 pruning is not verifier overfitting: acceptance needs only one pass and no policy or grounding failure, 85 of 311 accepted bundles fail some verifier case, and every non-dead pruned rule (96 of 127; Gemini 13 of 13) fires on a verifier case and fails it. The 127 extra rules add no hidden success: 31 are dead, matching no verifier or hidden case, and the rest fire on 242 hidden task exposures with 0 automated passes (114 incorrect escalations, 92 incorrect approval requests, 26 functional failures, and 10 abstentions, all policy-safe), which Safe-Subset turns into no-rule-match, so VAC and SC-VAC are unchanged in all 112 pruned units. These are post-hoc, zero-model-call, DSL-level counts, not production latency; Appendix E prices the stress arms.

Evidence timing does not replicate. The preregistered evidence-delay hypothesis (H3), the gap-3 minus gap-0 Direct raw-AUC effect, is -0.011 [-0.076, +0.045], failing to replicate v0.2's -0.090 penalty; the Gemini arm instead shows $+0.095$ [0.067, 0.123], so the model-stratified pooled $+0.042$ [0.011, 0.068] is driven entirely by Gemini, and the point estimates differ in sign across arms. Post-hoc, the difference is confined to the 4 access organizations (DeepSeek -0.085, Gemini +0.122; expense +0.063 versus +0.069), and model is confounded with thinking level and serving stack, so we make neither a domain nor a model claim. Incomplete authorization did not consistently induce abstention and coincided with unsafe action in some access cases: presumed model conservatism is not a release control.

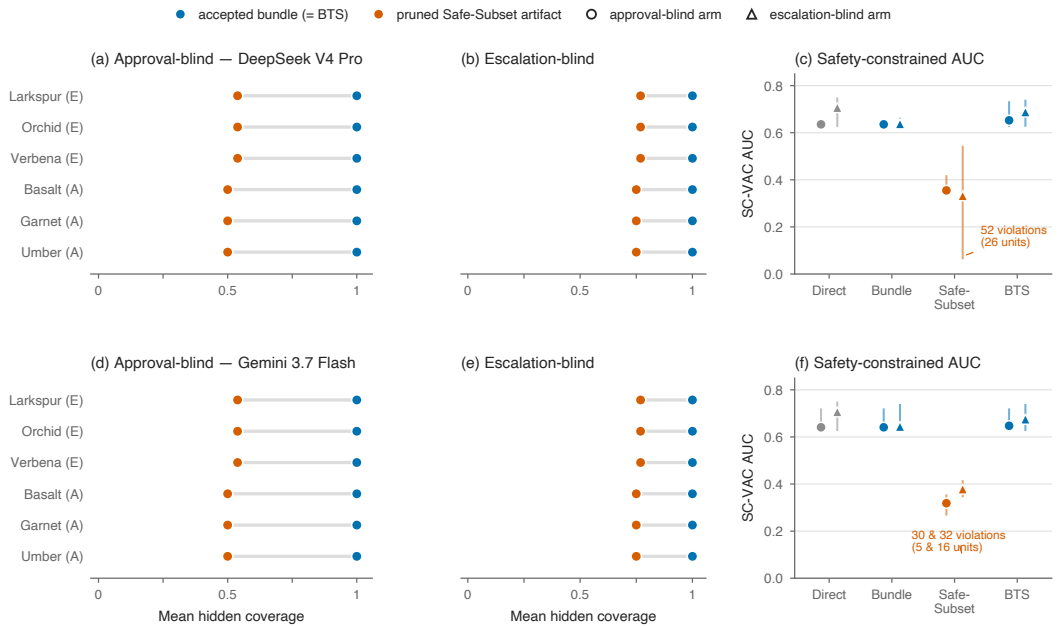


Figure 2: The two designed-blind-spot stress arms (top: DeepSeek; bottom: Gemini; six shared organizations). (a, b, d, e) Mean hidden VAC per organization over the nine evidence-complete units: the accepted bundle, which BTS retains, versus the pruned Safe-Subset artifact; each gap equals that organization's blind-task share. (c, f) Safety-constrained AUC by gate; markers are cluster means (circles approval-blind, triangles escalation-blind), whiskers span organizations.

5 Implications and related work

The over-pruning is caused by the fewer-rules tie-break, not by subset search, so an accepted artifact should not be minimized with a size preference. Releasing it unchanged (BTS) or breaking ties toward more rules both avoid the failure, the latter equally well or better under the full verifier (Section 4); BTS's remaining edge is fewer verifier evaluations, and on the rejected path a descending or greedy recovery, tested under the full verifier only, matches or beats its frozen fewer-rules one with fewer evaluations (Appendix E). Where verifier false acceptances are likely—thin evidence, or a panel lacking a decisive approval case—BTS inherits them while pruning may incidentally remove them; there the verifier, not the release operator, is the control to strengthen. The principle needs only a decomposable artifact and a verifier blind to some of its components, so it should transfer to tool-use plans, routing tables, or skill libraries with separable parts; where components interact, so that deleting one changes the verifier-visible behavior of others (as in free-text prompts), Proposition 1 does not apply.

Evidence timing yields no transferable monotonic rule: systems should explicitly detect incomplete threshold–authorization pairs rather than assume that the model will abstain. SC-VAC is deliberately artifact-level—one observed policy or grounding failure makes a checkpoint undeployable—so BTS's SC-AUC advantage over Direct is not raw-capability dominance; Table 1 reports the raw loss and violation reduction side by side.

Our over-pruning result is candidate-relative regression within one rule-decomposable proposal, not the harness-level forgetting that HCL's finite retention anchors cannot rule out [11].

Finite-test selection is known to admit solutions that fail independently evaluated behavior in automated program repair [17], where most patches reported by early generate-and-validate systems were incorrect and equivalent to deleting functionality [15]; our hidden scenarios serve this independent-evaluation role. Test-based debloating shows the same cost: removing never-executed code breaks needed functionality, which RAZOR counters with heuristics that retain related code [16] (BTS instead keeps the whole accepted artifact rather than inferring which unexercised parts matter), and dynamic-analysis-based debloaters remove up to 94% of code that should be retained

[1]. The designed blind spots are test-adequacy probes in the spirit of mutation testing, which seeds deliberate faults to measure what a test suite cannot see [3]. Safe policy improvement constrains deviation from a deployed baseline [13], and Seldonian and high-confidence policy-improvement algorithms select a candidate on one data split, safety-test it on another, and otherwise return No Solution Found [19, 20]; BTS's accept/recover/abstain branches share that shape without a statistical guarantee. Cho and Sun give an always-valid rule for when an iterative generate–verify workflow should release, calibrating evaluator scores against hard-negative failures and accumulating evidence with an e-process [2]; we instead ask what to release once a finite verifier has accepted a candidate.

Continual Harness edits prompts, sub-agents, skills, and memory online and studies model–harness co-learning [12]; we freeze the model and compare release operators. ExpeL and Agent Workflow Memory externalize reusable experience [21, 26]; ADAS and harness-improvement work search agent programs or task harnesses [7, 10, 27]. WorkArena, τ-bench, and CRMArena test agents in fixed enterprise environments and MiniCorp in a simulated company [4, 8, 22, 24]. Our object is complementary: chronological organization acquisition under an external release controller.

Artifact availability. Hidden panels are disposable by design (each benchmark version was frozen with a fresh hidden split), so with all arms run the retired panels are released in full at `https://github.com/gloriaowowo/bundle-then-subset`: code and hidden-case constructors (Apache-2.0), and frozen specifications, freeze hashes, prompts, stored candidates, run records, and every post-hoc analysis (CC-BY-4.0). Its commands re-execute all 12,480 stored hidden gate evaluations with zero model calls and re-derive the outputs behind every reported number, requiring exact matches (Appendix C). Credentials and raw provider responses are not released.

6 Limitations and responsible use

Organizations and tools are synthetic: eight parameterized organizations from two task-family templates in the confirmation and six in the stress arms, generated workflows of two to six rules (at most eight allowed), and verifier and hidden cases drawn from one typed specification per organization. That shared specification, together with coarse binary accept and deploy labels, likely explains why the full verifier separates every bundle in both confirmatory arms. All gates reach final VAC 1.000 under DeepSeek (0.990 under Gemini), and the evidence covers two temperature-zero models. The Gemini arm differs from the DeepSeek arm in thinking level and serving stack as well as in model, so the replication does not isolate the model factor, and two models do not establish generality across providers, scales, or sampling temperatures. The study therefore concerns acquisition trajectories under a small exact-search regime, not terminal capability or production-scale search cost. The cluster sample comprises these eight organizations; provider repetitions and bootstrap draws do not create additional organizations. The evidence intervention moves one policy and excludes noisy, adversarial, private, or continuously changing evidence. The v0.5 and v0.5b blind spots are designed, so they demonstrate a mechanism rather than estimating how often deployed verifiers are blind; undesigned rules that match no verifier case do occur in v0.3 (in 31 of 311 accepted DeepSeek bundles, none of Gemini's 300), but all are dead rules that match no hidden case either. The stress-split hidden panels are not comparable to v0.3's, and each model's two stress arms share cached candidates (a disclosed paired design, not independent samples). We leave to future work an adapted τ-bench case (policy clause $\rightarrow$ conditional rule, API $\rightarrow$ declared tool, database-state check $\rightarrow$ verifier or hidden case); composition with a stateful updater that carries released artifacts forward; a release holdout separate from the verifier and hidden panels; coverage-preserving minimizers and recovery search where exhaustive enumeration is infeasible (BTS's accepted path needs none; Appendix E); and non-rule artifacts such as prompts, routing policies, or tool-use plans.

Most importantly, BTS inherits accepted-bundle verifier errors by construction; its conservatism protects hidden-useful rules, not against false acceptance, just as resampling against an imperfect verifier cannot lower its false-positive probability [18]. v0.3 is the preregistered confirmation, replicated across both models, and the v0.5-family arms are prospective frozen stress tests; v0.2 BTS and every other analysis here, including all verifier-mask, Max-Subset, recovery-variant, cost, rescue, and organization- or domain-split results, are post-hoc diagnostics.

Production additionally requires access control, privacy review, human policy ownership, monitoring, rollback, and adversarial-evidence handling. We keep authorization, deterministic tests, and promotion outside the editable layer and make no production-readiness claim.

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A Freeze and audit

Both phases retain immutable run sets and aggregate-only hidden artifacts. The v0.3 freeze pins the benchmark, the implementation and analysis code, and the prompt contract by cryptographic hash; the exact digests, together with every preserved v0.2 record, are part of the released archive rather than reproduced here.

The audit verified 480 unique v0.3 units, all manifests/artifacts, all frozen v0.3 hashes, every preserved v0.2 hash, zero infrastructure failures, exhaustive mutually exclusive outcome counts, and no hidden task-level IDs, traces, grades, or failure strings.

The post-hoc v0.4 analysis preserves every frozen hash, makes zero model calls, reproduces all full-verifier gate-unit results, and then enumerates 255 masks. The v0.5 stress arm carries its own frozen protocol, freeze record, zero-provider oracle validation, and audited 90-unit run set under the same contract. Its three predictions and full protocol were frozen before the hidden matrix ran, after a two-organization development check and after the zero-provider oracle validation had already established the mechanism at the oracle level; its evidence-complete stratum was named only later, in the v0.5b protocol. In v0.5 the evidence-complete coverage cost is 0.46 in the expense and 0.50 in the access organizations; the three access organizations coincide because their panels share size and approval share. Because v0.5b's organizations, evidence, prompts, and model configuration are byte-identical to v0.5, the deterministic response cache resolves v0.5b generations to the v0.5 responses, pairing 68 of the 72 candidates (86 of 90 units). The exception is one seed series of one expense organization (four units) whose first v0.5 call terminated in each unit and was retried as the second permitted attempt; v0.5b regenerated them fresh in one attempt each, for all of its $0.006 provider cost, so the frozen protocol's statement that the candidate hashes match holds for 86 of 90 units, and per-arm counts are computed on each arm's own candidates. The pairing is disclosed in the frozen protocol, and the v0.5b predictions were frozen before any v0.5b evaluation was computed. The v0.2 BTS reconstruction and all v0.4 analyses are labeled post-hoc and do not alter preregistered v0.3 H1–H3. Two preregistered mandatory outputs complete the v0.3 record: Direct's gap-1 minus gap-0 raw AUC is +0.070 [+0.027, +0.107], so the evidence-delay curve is not monotonic, and the weight λ above which BTS's normalized functional AUC minus λ times violation-exposure AUC exceeds Direct's lies between 0.42 and 0.74 across the four gaps. Post-hoc, the model-by-gap interaction in H3 is +0.107 (t-interval [+0.010, +0.203] over 8 organizations instantiating two templates; descriptive, not a test); the gap effect is negative under DeepSeek and positive under Gemini in 4 of 4 access and 0 of 4 expense organizations.

Relative to the reviewed version, this camera-ready adds post-hoc, zero-model-call analyses of stored candidates: the Max-Subset tie-break control and its per-mask replay, cross-model accept/reject concordance, full-verifier separation in the Gemini arm, per-mask verifier-ablation distributions (Appendix D), verifier-work, early-exit, retained-rule, pruned-rule, and recovery-variant accounting (Appendix E), and the v0.2 domain and H3 organization splits. None alters a preregistered result.

B Cross-model successor protocols

The Gemini arm executed under two successively frozen successor protocols, each recording a forced infrastructure deviation while inheriting every scientific element byte-identically. First, the provider rejects the frozen reasoning-off configuration for this model (an HTTP 400 response stating the minimal thinking level is unsupported for this model), so the arm runs at the lowest supported thinking level; reasoning level is part of the serving configuration, which is why the replication is reported as cross-model *and* cross-configuration. A second configuration difference surfaced only after review: although the frozen model descriptor sets JSON mode for both arms, the Vertex adapter of the pinned client library does not forward that setting, so Gemini's structured output was enforced by the prompt and the schema-validating parser (which strips code fences) rather than by the provider. Second, the consumer Generative Language API stalled on provider-side capacity—sustained HTTP 503 responses degrading to timeout-length hangs, with 48 of 480 hidden units complete at the pre-stated decision point after a day of retries—so execution moved to the same model behind Vertex AI under a decision rule stated in advance; all 480 units then ran fresh on the new route, and the abandoned partial evidence is archived unpooled. Third, development on the new route surfaced candidates that pass schema validation but fail deterministic execution (for example,

binding a non-numeric value to a numeric tool argument, which this model occasionally does and DeepSeek never did); a guard adopted before the freeze classifies such candidates as rejected generations; it fired on two of 32 eligible development units and zero of 384 eligible hidden units (the single rejected hidden generation exhausted its attempts without a valid structured output). Provider-capacity errors (HTTP 503/429) were classified as infrastructure failures and absorbed by the frozen retry rules; no scientific outcome was derived from a failed attempt. The arm's audit verified all 480 units, manifests, frozen hashes, and outcome counts with no hidden task-level retention. Billable cost was $2.10 summed over every archived v0.3c run record, including an aborted first development launch and infrastructure retries. The successor freeze records, the transport-route protocol, and the guard's audit trail are part of the released archive.

The Gemini stress arms (v0.5g, v0.5bg) ran under jointly frozen successor protocols with the same benchmarks, panels, gates, and analysis, the Vertex transport route and execution-validity guard of the cross-model arm, and candidates fully cache-paired within the model ($90/90$ units). Every Gemini candidate encodes the blind obligation ($72/72$ in both arms), subset search prunes the blind rule from every accepted bundle ($66/66$ approval-blind, $57/57$ escalation-blind), and the coverage loss reproduces the DeepSeek cluster means to all reported digits (0.48 and 0.24). In the escalation-blind arm 16 of 54 evidence-complete units relied on escalate-first ordering (32 Safe-Subset violations, against DeepSeek's 52 across 26 units); the arm's freeze record carries v0.5b's full P3', including the benign-shape point prediction at evidence-complete checkpoints (the jointly frozen successor protocol prose states only the disjunction, benign abstention or rule-order violations); the point prediction failed (16 of 54 units violating), as it failed under DeepSeek in 26 of 54 units, and the disjunction held. Unlike DeepSeek, whose approval-blind arm was fully benign, 5 of 54 approval-blind units relied on approval-rule ordering (30 Safe-Subset violations), so the transferred P3 point prediction of a benign shape fails for Gemini's approval-blind arm, and the observed shape is reported per the frozen commitment. Counting against the freeze records, the benign-shape point prediction failed in three of the four arms, all but DeepSeek's approval-blind arm. The clean arm-determined flip is therefore DeepSeek-specific.

C Reproduction, compute, and assets

The released archive (`https://github.com/gloriaowowo/bundle-then-subset`) licenses code—the simulator, workflow DSL, gates, grader, hidden-case constructors, and analysis and audit scripts—under Apache-2.0, and specifications, protocols, freeze records, run records, analyses, and documentation under CC-BY-4.0. The hidden panels are disposable by design: each benchmark version (v0.2, v0.3, v0.5) was frozen on newly authored organizations with a fresh hidden split, and the cross-model (v0.3c, v0.5g, v0.5bg) and escalation-blind (v0.5b) arms reused their parent's organizations by prespecified design. All arms have run, so the retired v0.2–v0.5 panels are released in full: the four distinct benchmark constructors, their full frozen specifications, which re-hash to the frozen benchmark hashes, and all 2,539 archived run units with their stored candidates. Any stored candidate can therefore be re-evaluated on its hidden panel with zero model calls. Because the v0.5 organizations reuse the v0.3 evidence template under new skins, a future confirmation needs a new template, not only new organizations. A dataset card documents construction, schemas, intended research use, and limitations, and a contamination canary string marks the release documentation and the data files written by the release builder; analysis outputs and hash-frozen files stay byte-identical and are covered by the release manifest instead. Seven lines of non-scientific session wording in five hash-frozen protocol and freeze records are replaced in the release copies; an erratum lists each edited line, each edited file's frozen and released SHA-256, and a SHA-256 of each removed line.

Every command below makes zero model calls; none needs the network once the pinned figure environment is installed (a one-time `pip install`). `node artifact_tools/audit-and-reproduce.mjs` re-derives every reported aggregate summary (v0.2, v0.3, v0.3c, and the four stress arms) from the run records with exact equality, recomputes the paper's result macros, and hash-verifies every archived run record. `node scripts/release-camera-ready.mjs` reruns the six camera-ready analyses that need only stored run records (verifier work, full-verifier separation, cross-model concordance, stress-arm cache pairing, v0.2 pruning by domain, and H3 by organization), regenerates the camera-ready macros and tables and Figures 1 and 3, and checks every camera-ready analysis output against frozen SHA-256 hashes; with `--with-hidden` it also reruns, on the released hidden panels,

the seven camera-ready analyses that re-evaluate hidden cases—the verifier-mask replay, the retained-rule accounting, the Max-Subset control, the recovery variants, the early-exit evaluation counts, the Max-Subset mask replay, and the pruned-rule verifier behavior—before the hash check. `node artifact_tools/rerun-hidden.mjs` validates every constructor (oracle workflows on every v0.3–v0.5 development and hidden case; panel and gate checks for v0.2); re-executes all 12,480 stored hidden gate evaluations of the seven hidden runs (v0.2, v0.3, v0.3c, and the four stress arms) on their hidden panels and requires each rebuilt record to equal the stored one; re-aggregates the four stress-arm hidden summaries from those records; and reruns the v0.4 verifier ablation and retention replay with their audits, requiring byte-identical outputs. `node artifact_tools/verify-release.mjs` checks every file against the release manifest, recomputes or erratum-attests all 121 hashes pinned by the eight freeze records, and reruns the hidden-run audits, which reproduce the released audit records byte for byte. `node toy/run-toy.mjs` runs the public toy benchmark.

Not released are credentials and raw provider responses (the provider-response cache; candidates are released as parsed workflows), so the original paid candidate generation cannot be replayed offline; regenerating candidates needs provider keys. Task-level traces were never persisted in any split; replaying stored candidates regenerates them. The v0.2 run-state file was not archived, so the frozen v0.2 audit script cannot rerun; its pinned hashes and run set are checked instead. Exploratory pilots and connectivity probes outside the archived run sets required additional provider calls; they support no reported result and are not part of the release.

The provider does not disclose serving hardware. The 480-unit v0.3 run had an observed asynchronous timestamp span of 23.4 minutes and cost $0.5082. The 90-unit v0.5 stress arm used the same DeepSeek configuration and cost $0.128 including its development stage, with zero infrastructure failures as classified by its frozen audit (the four first calls that terminated and were retried count as generation attempts; Appendix A); the paired v0.5b arm cost $0.006, all for regenerating the one seed series that the cache did not pair (Appendix A). The 480-unit Gemini arm cost $2.10 summed over every archived v0.3c run record (development, hidden, an aborted first development launch, and infrastructure retries), and the Gemini stress arms cost $0.48 (approval-blind, including development on both transport routes) and $0.00 (escalation-blind, fully cache-paired); provider-capacity errors (HTTP 503/429) were classified as infrastructure failures and absorbed by the frozen retry rules, and no scientific outcome was derived from a failed attempt. Deterministic validation and analysis used an 11-core Apple M3 Pro CPU with 18 GB RAM, no GPU, and under 0.2 GB local artifact storage; a v0.4 precomputed-output integrity audit completes in well under a second on that machine. The Max-Subset tie-break control is a zero-model-call replay of stored candidates through the frozen gates; its script and output are packaged in the archive. An earlier copy of that output summed the four evidence regimes of the confirmatory arms (a series-key bug); the values reported here were unaffected, and the released output is corrected.

The implementation uses Pi Agent Core and Pi AI v0.84.2 under the MIT License and TypeScript v7.0.2 under Apache-2.0. DeepSeek V4 Pro was accessed under the provider's API terms on August 22 (v0.3) and August 26 (v0.5), 2026; Gemini 3.7 Flash was accessed under Google's API terms via the Generative Language API and Vertex AI on August 26, 2026. The organizations and task instances are synthetic assets authored for this study; the released archive documents their construction, schemas, intended research use, and limitations, with the licenses above and third-party notices.

D Per-mask verifier ablation

This analysis is post-hoc, makes zero model calls, and covers the v0.3 DeepSeek arm only. The replay re-evaluates every stored candidate under each of the 255 nonempty subsets of the eight-case verifier (254 weakened panels plus the full panel) and reproduces the committed aggregate ablation; each mask is scored by its mean AUC over 96 seed series. Table 3 counts masks by the sign of the BTS-minus-Safe-Subset difference, and Figure 3 pairs each mask's coverage and violation changes. Averages favor BTS: across the eight leave-one-case-out masks, BTS retains 0.697 SC-AUC with 0.00086 violation AUC, versus Safe-Subset's 0.633 and 0.00927; as one, two, then three of the six semantic strata are dropped (6, 15, and 20 masks), Safe-Subset's mean SC-AUC falls $0.566 \rightarrow 0.425 \rightarrow 0.301$ while BTS's declines only $0.685 \rightarrow 0.665 \rightarrow 0.646$ and Bundle $0.628 \rightarrow 0.627 \rightarrow 0.625$, and Safe-Subset's mean violation exposure is 3.46, 3.29, and 2.89

Table 3: **Post-hoc, zero model calls; v0.3 DeepSeek only.** Per-mask comparison of BTS with Safe-Subset over the replayed verifier subsets. Each mask's metric is the mean AUC over its 96 seed series; "lower/equal/higher" counts masks by the sign of BTS minus Safe-Subset violation-exposure AUC (ties within 10^{-12}). The first block keeps whole semantic strata (Fig. 1(c) grouping); the full panel is the single six-strata mask and is included in both "all" rows. D3 is the special-approval case. Mean ΔSC-AUC is never negative for any mask, but BTS has lower SC-AUC than Safe-Subset in 140 of 24,480 individual seed series. The violation sign depends mainly on the escalation case D8 and, when D8 is present, on D3: BTS is never lower with D8 and is higher in every D8 mask that omits D3; of the 76 masks where BTS is higher, 64 include D8 and omit D3.

		BTS violation AUC vs. Safe-Subset				
Mask group	Masks	Lower	Equal	Higher	Mean ΔViol.	Mean ΔSC
1 of 6 semantic strata	6	2	2	2	−0.0094	+0.453
2 of 6 semantic strata	15	7	2	6	−0.0245	+0.417
3 of 6 semantic strata	20	9	3	8	−0.0275	+0.344
4 of 6 semantic strata	15	5	5	5	−0.0216	+0.240
5 of 6 semantic strata	6	1	4	1	−0.0112	+0.118
6 of 6 semantic strata	1	0	1	0	0.0000	0.000
All semantic-strata masks	63	24	17	22	−0.0217	+0.320
All case subsets	255	112	67	76	−0.0279	+0.284
D8 (escalation) absent	127	112	11	4	−0.0599	+0.460
D8 (escalation) present	128	0	56	72	+0.0039	+0.108
D8 present, D3 absent	64	0	0	64	+0.0064	+0.114
D8 present, D3 present	64	0	56	8	+0.0013	+0.102

times BTS's. Four facts qualify these averages. First, BTS's coverage is never below Safe-Subset's per mask or per organization, but can be below it at finer levels: its mean SC-AUC is below Safe-Subset's in no mask and in none of the 2,040 organization–mask pairs, but it is lower in 140 of 24,480 seed series and 1,366 of 122,400 unit–mask cells. Second, the violation comparison is close to even over the semantic-strata masks of Figure 1(c) (24/17/22 lower/equal/higher) and depends mainly on the escalation case D8 and, when D8 is present, on the special-approval case D3: BTS is lower/equal/higher in 112/11/4 of the 127 masks without D8 and in 0/56/72 of the 128 with it. Of the 76 masks where BTS is higher, 64 include D8 and omit D3 (all 64 such masks), where Bundle falsely accepts and BTS inherits the error; 8 include both (of 64 such masks), and 4 omit D8. Third, against Bundle, BTS's violation exposure is never lower and is higher in 98 of 255 masks (0/157/98), the price of recovering subsets with a weakened verifier. Fourth, BTS's coverage edge over Safe-Subset is a tie-break effect. Replayed on the same masks, Max-Subset's mean SC-AUC is higher/equal/lower than BTS's in 168/66/21 of 255 masks (never lower by more than 0.0009) and never below Safe-Subset's (234/21/0); its violation exposure is lower/equal/higher than BTS's in 98/117/40 masks. Over the 254 weakened panels its mean SC-AUC is 0.672 (BTS 0.660, Bundle 0.623, Safe-Subset 0.375) and its mean violation AUC 0.0071 (BTS 0.0083, Bundle 0.0068, Safe-Subset 0.0363). BTS with a more-rules recovery releases exactly Max-Subset's artifact in every unit of all 255 masks. The masks are exhaustive subsets of one synthetic panel, not a sample of real verifiers.

E Verifier work, recovery variants, and the cost of retained rules

All analyses here are post-hoc and make zero model calls: verifier work is counted from stored run records, and retained-rule behavior from a replay of stored candidates through the frozen simulator and grader. Safe-Subset evaluates $2^n - 1$ artifacts per n-rule candidate; BTS evaluates one if the bundle is accepted and $1 + (2^n - 1)$ otherwise. With full-bundle acceptance rate a among n-rule candidates the reduction is therefore $a - 1/(2^n - 1)$, so it concentrates where acceptance is high (Table 5): 15.1% for three-rule v0.3 candidates but 98.4% for six-rule ones, 88.6% overall, and 81.5% in the Gemini arm (952 versus 5,133). These savings assume that Safe-Subset is implemented, as here, by plain enumeration of all $2^n - 1$ subsets. A pass-maximizing search that ties toward more rules could stop after testing the complete bundle whenever it passes every verifier case, as 226 of

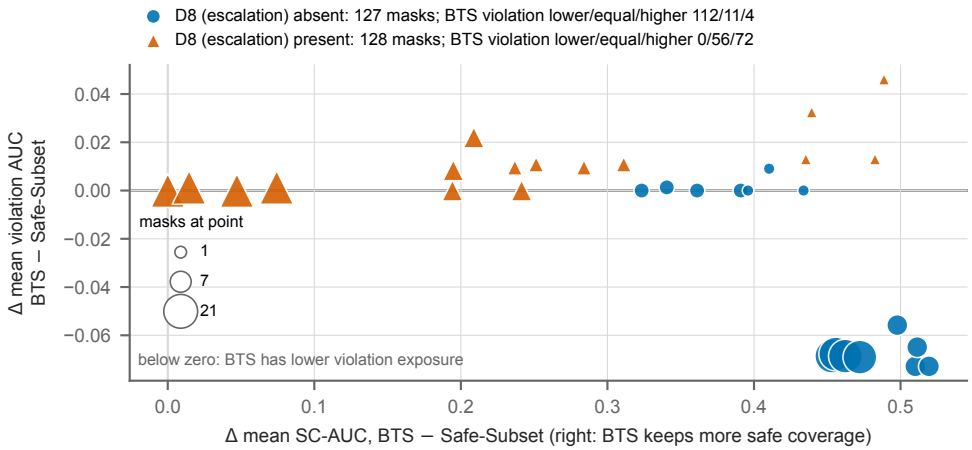


Figure 3: **Post-hoc, zero model calls; v0.3 DeepSeek only.** Each point is a verifier mask: BTS minus Safe-Subset mean SC-AUC (horizontal) against mean violation AUC (vertical); marker area counts masks sharing a point. Color and shape mark whether the escalation case D8 is in the panel; without it BTS has lower violation exposure in most masks, with it never.

311 accepted DeepSeek and 251 of 300 accepted Gemini bundles do; for the other accepted bundles the simple variant counted here enumerates every subset, whereas BTS stops at acceptance. Counted on the stored records, this early-exit Max-Subset needs 2,159 (DeepSeek) and 1,171 (Gemini) evaluations, so BTS's saving over it is 50.8% and 18.7% rather than 88.6% and 81.5%; a search that descends by subset size and stops at the first subset passing every case could need fewer, so these are upper bounds on BTS's saving over early-exit search. On the rejected path BTS pays whatever its recovery search costs, so for a rule-count group that is always rejected it pays more than Safe-Subset (Gemini's two-rule candidates: 12 versus 9).

Table 6 prices retention in every arm. In the confirmatory arms the rules that Bundle and BTS keep but Safe-Subset prunes never produce an automated hidden pass, and none of them fires before a rule Safe-Subset kept on any of the 3,889 (DeepSeek) and 3,757 (Gemini) hidden tasks of accepted units; Gemini keeps only 13 extra rules and has no dead rules. In the stress arms the retained rules carry the blind behavior; they also fire before a rule Safe-Subset kept on 79 of 701 (DeepSeek) and 62 of 714 (Gemini) escalation-blind hidden tasks and on 30 of 822 in Gemini's approval-blind arm. Whether a misrouted escalation costs more than an abstention is a deployment judgment. Latency, memory, and interference beyond first-match overlap are not measured.

Table 4 reports two cheaper recovery searches, designed post-hoc after the Max-Subset control and evaluated only with the full eight-case verifier, never under weakened masks; they are exploratory. Descent tries subsets from the largest size down and keeps the most-passing safe subset of the first size that has one (remaining ties go to the first in enumeration order); greedy drops one rule at a time. Both need fewer verifier evaluations than frozen BTS because they do not enumerate every subset (59% and 66% of BTS's under descent in the confirmatory arms), and both match or exceed its SC-AUC in every arm because they rescue larger subsets (mean 2.30 versus 1.74 rules in DeepSeek v0.3) and so avoid the fewer-rules tie-break on the rejected path. Frozen BTS is therefore not the best point in this design space, and the 88.6% saving over plain enumeration is not the best attainable figure either.

Larger search spaces. Exhaustive enumeration grows as 2^n, but BTS searches only on its rejected path: an accepted bundle costs one verifier evaluation whatever n is, so with full-bundle acceptance rate a the expected work per n-rule candidate is $1 + (1 - a)\, R(n)$, where $R(n)$ is the mean number of evaluations the recovery search spends on a rejected candidate ($2^n - 1$ for exhaustive search, which re-tests the bundle). Where enumeration is infeasible, a cheaper search can fill that role: greedy deletion needs $O(n^2)$ evaluations, descending subset sizes $O(n^d)$ when d rules must be dropped (exponential in the worst case), and delta debugging is another option. By the argument of Proposition 1, which does not use acceptance, any search run to 1-minimality (as delta debugging is) removes every rule that no verifier case matches, because deleting such a rule never changes a verifier

outcome. Descend avoids this by construction, since it returns a largest passing subset; greedy stops at the first passing subset on its deletion path, which keeps such rules unless a tied deletion earlier on the path removed one. In v0.3 they matched exhaustive SC-AUC with 59% (descend) and 58% (greedy) of its evaluations under DeepSeek and 66% under Gemini (Table 4). These variants are post-hoc and were evaluated only under the full verifier, and their searches ran only on rejected candidates of at most five rules, most of them three; behavior with larger artifacts, weakened verifier masks, or stochastic and expensive verifiers is untested.

Table 4: **Post-hoc, zero model calls; designed after the Max-Subset control.** Recovery variants of BTS: each releases an accepted bundle unchanged and differs only in its search after rejection. Exhaustive (frozen BTS) enumerates all subsets with ties toward fewer rules; *descend* tries subset sizes from $n - 1$ downward and stops at the first size with a safe subset that passes a case; *greedy* deletes one rule at a time. Cells give SC-AUC / total verifier artifact evaluations under the full verifier.

	v0.3 confirmation		Approval-blind		Escalation-blind	
Recovery	DeepSeek	Gemini	DeepSeek	Gemini	DeepSeek	Gemini
Exhaustive (BTS)	0.704 / 1,063	0.705 / 952	0.653 / 122	0.647 / 114	0.688 / 200	0.675 / 177
Descend	0.704 / 626	0.705 / 629	0.657 / 91	0.647 / 90	0.732 / 131	0.707 / 135
Greedy	0.704 / 620	0.705 / 629	0.657 / 91	0.647 / 90	0.732 / 128	0.707 / 129

Table 5: **Post-hoc accounting from stored run records, zero model calls.** Verifier artifact evaluations by candidate rule count n. Safe-Subset tests all $2^n - 1$ nonempty subsets of every candidate; BTS tests the complete bundle once and, only if it is rejected, runs the same exhaustive search. Within a rule-count group with bundle acceptance rate a the reduction is exactly $a - 1/(2^n - 1)$, so the saving comes from large, usually accepted candidates. Skipping BTS's redundant re-test of the rejected full bundle would give 993 (89.3%) in v0.3. Across all arms, generated workflows contain two to six rules.

	Candidates	Accept rate a	Safe-Subset evals	BTS evals	Reduction
v0.3 DeepSeek, $n = 3$	68	29.4%	476	404	15.1%
v0.3 DeepSeek, $n = 4$	112	81.3%	1,680	427	74.6%
v0.3 DeepSeek, $n = 5$	172	99.4%	5,332	203	96.2%
v0.3 DeepSeek, $n = 6$	29	100.0%	1,827	29	98.4%
v0.3 DeepSeek, all	381	81.6%	9,315	1,063	88.6%
v0.3c Gemini, all	383	78.3%	5,133	952	81.5%

Table 6: **Post-hoc replay of stored candidates, zero model calls.** Cost of keeping the accepted bundle, on accepted units only, where Bundle and BTS release the bundle and Safe-Subset releases its pruned subset; per-task quantities are computed on the same units for both gates (unit-weighted, except multi-match shares, which are pooled over tasks). A pre-emption is a hidden task on which a retained rule fires before a rule Safe-Subset kept. In v0.3 DeepSeek the 242 firings of retained rules are 114 incorrect escalations, 92 incorrect approval requests, 26 functional failures and 10 explicit abstentions, all policy-safe; Safe-Subset turns them into no-rule-match, and VAC and SC-VAC are unchanged in all 112 pruned units. Dead rules match no verifier case and no hidden case even when executed alone. Latency and memory are not measured.

	Confirmatory		DeepSeek stress		Gemini stress	
	DeepSeek	Gemini	Appr.-blind	Esc.-blind	Appr.-blind	Esc.-blind
Accepted units	311	300	66	56	66	57
Rules released, Bundle/BTS	1,453	1,180	327	297	258	231
Rules released, Safe-Subset	1,326	1,167	174	212	147	171
Extra rules kept	+9.6%	+1.1%	+87.9%	+40.1%	+75.5%	+35.1%
Units Safe-Subset prunes	112	13	66	56	66	57
Cond. evals/task, Bundle/BTS	2.612	2.627	2.994	2.458	2.944	2.693
Cond. evals/task, Safe-Subset	2.536	2.607	2.308	2.143	2.031	2.309
Tasks with > 1 match, Bundle/BTS	12.0%	9.6%	3.8%	11.3%	9.2%	15.3%
Tasks with > 1 match, Safe-Subset	10.4%	9.3%	3.0%	0.0%	1.9%	6.6%
Pre-emptions of a kept rule	0	0	0	79	30	62
Hidden firings of retained rules	242	59	451	176	447	183
automated passes	0	0	372	168	372	171
policy violations	0	0	20	0	18	0
policy-safe non-passes	242	59	59	8	57	12
Dead rules in accepted bundles	31	0	26	27	0	0

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    Answer: [Yes]

    Justification: Sections 2–4 describe the frozen generators (DeepSeek V4 Pro reasoning-off and Gemini 3.7 Flash at the lowest supported thinking level), including their proposal-only role, temperature-zero setting, bounded static repair, and lack of behavioral feedback.

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